

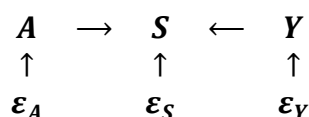
Causal Inference

Quiz Date: 2025/10/17

Quiz Time: 1.5 hours

Dept/Year: 統計三、統計四

1. It seems that every time I attend class, the professor forgets to take attendance. Yet, on the one occasion I was absent, he remembered to do so. Clearly, my presence in class causes the professor to forget!
 - a. Identify the type of fallacy. (5%)
 - b. Explain why this reasoning is flawed. (5%)
2. For each individual unit in a study, can we observe both counterfactual outcomes $Y_i(1)$ and $Y_i(0)$?
 - a. Answer yes or no. (5%)
 - b. Briefly explain your answer. (5%)
3. Consider the **DAG**,



Assume a Nonparametric Structural Equation Model (**NPSEM**) with mutually independent exogenous errors $\varepsilon_A, \varepsilon_S, \varepsilon_Y$.

- a. Write down a simple **NPSEM** that corresponds to this **DAG**. (5%)
 - b. If we control on $S = s$, does it hold that $A \perp Y|S$? Answer yes or no and show your conclusions using **NPSEM**. (10%)
4. Let

$$Y = \begin{cases} 1, & \text{if student passes the final} \\ 0, & \text{if student fails the final} \end{cases} \quad A = \begin{cases} 1, & \text{student attend tutoring} \\ 0, & \text{otherwise} \end{cases}$$

Define each student type by their counterfactual outcomes:

$$\left\{ \begin{array}{l} \text{Always - fail: } Y(0) = Y(1) = 0 \\ \text{Always - pass: } Y(0) = Y(1) = 1 \\ \text{Helped by tutoring: } Y(1) = 1, Y(0) = 0 \\ \text{Harmed by tutoring: } Y(1) = 0, Y(0) = 1 \end{array} \right.$$

Suppose you observe the following data for 5 students:

i	A_i	Y_i	$Y_i(1)$	$Y_i(0)$	Type
1	1	?	1	1	?
2	1	:	0	0	:
3	0	:	1	0	:
4	0	:	0	1	:
5	1	?	1	1	?

- a. Fill in the missing observed outcomes, assign each student a causal type. (5%)
- b. Compute

$$\text{Causal Effect} \triangleq E[Y(1)] - E[Y(0)]$$

$$\text{Association} \triangleq E(Y|A = 1) - E(Y|A = 0)$$

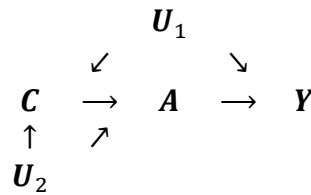
and compare the results. (10%)

5. Consider a linear model in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

What is condition for **OLS** estimator of $\boldsymbol{\beta}$ to be unbiased?

- State the key assumption about the error term $\boldsymbol{\varepsilon}$ and the independent variables $\mathbf{X}$ that is required for the **OLS** estimator of $\boldsymbol{\beta}$ to be unbiased. (5%)
 - Explain why violating this assumption would lead to biased **OLS** estimator. (10%)
6. Consider the **DAG**



- State the Backdoor Criterion. (10%)
 - Does confounding exist in this **DAG**? (10%)
7. A researcher is studying the relationship between study hours (A), exam scores (Y), and stress levels (C). To isolate the effect of study hours on exam score while controlling for stress, the following correlation matrix of (A, Y, C) were found:

	A	Y	C
A	1	17/25	4/5
Y		1	3/5
C			1

- Calculate the partial correlation between study hours and exam scores while controlling stress levels, which is $\rho_{AY|C}$. (10%)
 - Compare $\rho_{AY|C}$ to ρ_{AY} . Based on this comparison, what can you conclude about the role of stress in the relationship? (5%)
8. (Bonus)

In a Muay Thai match, Buakaw Banchamek is facing a rising star, Fighter X, in a major stadium fight. The bookmaker sets the following odds:

Buakaw to win: 3/2

Fighter X to win: 11/5

- Determine the bookmaker's margin for this market. (5%)
- What's the fair odds (margin= 0) would be? (5%)